Introductory Chemistry

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Introductory Chemistry (A06218)

Short Title:	Introductory Chemistry
Faculty	Science and Computing
Department	Science
Cluster	Chemistry
Author	MBREEN
Credits	5

Level: Introductory

Description of Module / Aims

This module aims to provide an introduction to the general principles of analytical and inorganic chemistry, and also to their applications. Topics covered will include stoichiometry, bonding and atomic structure, the periodic table and trends therein, acidity/basicity and reaction kinetics.

Programmes

CHEM-0001	BSc (Common Entry) (WD_SSCIE_D)	1 / 1 / M
CHEM-0001	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	1 / 1 / M
CHEM-0001	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	1 / 1 / M
CHEM-0001	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	1 / 1 / M
CHEM-0001	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	1 / 1 / M
CHEM-0001	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	1 / 1 / M
CHEM-0001	BSc (Hons) in Science (Common Entry) (WD_SSCCM_B)	1 / 1 / M
CHEM-0001	BSc in Food Science with Business (WD_SFDSC_D)	1 / 1 / M
CHEM-0001	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	1 / 1 / M
CHEM-0001	BSc in Pharmaceutical Science (WD_SPHSC_D)	1 / 1 / M

Indicative Content

- Matter and atomic structure
- Chemical reactions; the mole concept and stoichiometry; stoichiometric calculations
- Acid/base reactions; volumetric analysis; indicators; buffers, pH
- Oxidation/reduction reactions; balancing redox equations
- Electronic structure of the atom; ionic and covalent bonding; intermolecular forces
- Introduction to hybridisation and the shape of simple molecules
- Periodic table, periodic trends and chemistry of the elements
- Reaction kinetics

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe atomic structure, the factors that influence reaction rates, ionic/covalent/intermolecular bonding and acidity/basicity.
2. Calculate molar quantities, concentrations, reaction yields and theoretical pH.
3. Interpret chemical equations and trends/ properties associated with the periodic table.
4. Complete a titration and solve associated titrimetric calculations.
5. Communicate by written and verbal means the results of practical based exercises.
6. Demonstrate safe handling/disposal of chemicals and good laboratory practice (GLP).

Learning and Teaching Methods

- Lectures.
- Laboratory practicals.
- Online tutorials.
- Self-directed study.

Assessment Methods

	Weighing		Outcomes Assessed
Final Written Examination	40%		1,2,3,4
Continuous Assessment	60%		
Practical		40%	4,5,6
Tutorial/Problem Sheets		20%	1,2,3,4

Assessment Criteria

<40%: No understanding of the basic principles of analytical and inorganic chemistry. Inadequate practical skills in the associated laboratories.

40%-49%: Has a superficial knowledge of analytical and inorganic chemistry. Has a basic level of competency & skills in the associated laboratories.

50%-59%: Shows a reasonable level of understanding of the various principles involved in analytical and inorganic chemistry. Displays good practical skills in the associated laboratories.

60%-69%: Shows a good knowledge of the areas of analytical and inorganic chemistry, as well as an ability to reason and analyse problems related to the area. Has a high level of skills in the associated laboratories.

70%-100%: All the above to an excellent level, in addition to excellent reasoning skills and focused well thought out arguments. Engages in debate in class and has a high level of efficiency in the associated laboratories.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Tutorial	8	
Practical	16	
Independent Learning	75	

Essential Material

- "Mastering Chemistry." www.pearsonmylabandmastering.com
- Browne, T.L. and H.E. Lemay. _Chemistry: The Central Science_. 3rd ed. Australia: Pearson, 2013.

Supplementary Material

- Ebbing, D. and S. Gammon. _General Chemistry_. 11th ed. New York: Cengage Learning, 2016.
- Russo, S. and M. Silver. _Introductory Chemistry: Atoms First_. 5th ed.. San Francisso: Prentice Hall, 2010.
- Silbererg, M.S. and P.G. Amsteis. _Chemistry, The Molecular Nature of Matter and Change_. 7th ed. UK: McGraw-Hill, 2015.

Requested Resources

- Room Type: Chemistry Lab